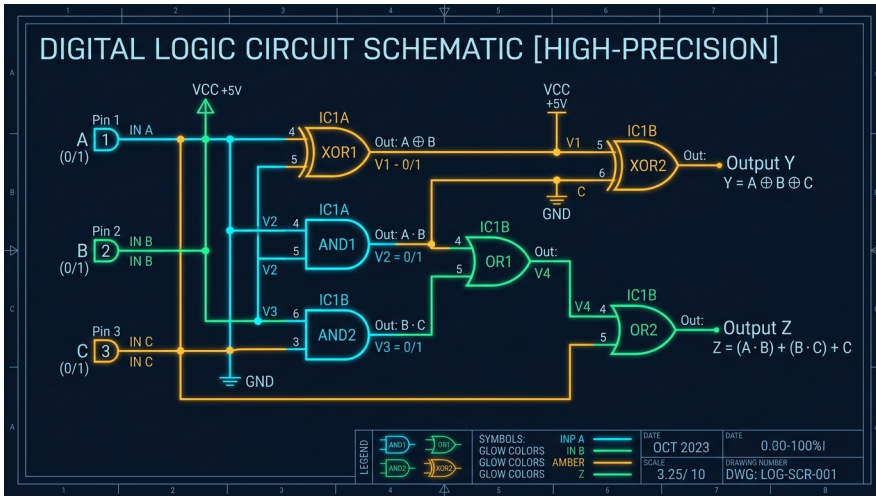


A Refresher on Boolean Logic

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“Nature proposes an ‘AND’. Our observation enforces an ‘OR’.”
— Professor Alvin

```
import schemdraw
import schemdraw.logic as logic
import schemdraw.elements as elm
```

Logical States

Before diving into the fascinating realm of quantum logic, mastering the fundamentals of classical logic and traditional logic gates is essential.

Logic gates are the foundational building blocks of digital electronics and modern computing. They manipulate binary signals—units of information represented by bits that take one of two distinct values: “0” (false, low, off) or “1” (true, high, on).

Throughout this blog, we will express all circuits in matrix form. So, we might as well get comfortable with this concept right from the start.

We define the logical states as:

$$\text{ZERO} = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

and

$$\text{ONE} = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

Logical Gates

Basic logic gates are the elementary building blocks of any digital system. These gates are: **NOT**, **AND**, **OR**, **NAND**, **NOR**, **XOR**, and **XNOR**. Each gate exhibits a specific behavior, formally defined by a truth table that maps its inputs to its output.

The NOT Gate (Inverter)

The NOT gate is the simplest of all logic gates. It takes a single input and returns its inverse.

Symbol :

Table 1: Truth Table: NOT Gate

Input (A)	Output (Y)
0	1
1	0

The output is the inverse of the input.

$$\text{NOT} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

The AND Gate

The output is 1 only if both inputs A and B are 1.

Table 2: Truth Table: AND Gate

Input A	Input B	Output Y
0	0	0
0	1	0
1	0	0
1	1	1

$$\text{AND} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The OR Gate

The output is 1 if at least one of the inputs is 1.

Table 3: Truth Table: OR Gate

Input A	Input B	Output Y
0	0	0
0	1	1
1	0	1
1	1	1

$$\text{OR} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}$$

The NAND Gate (NOT-AND)

The output is 0 only when both inputs are 1.

Table 4: Truth Table: NAND Gate

Input A	Input B	Output Y
0	0	1
0	1	1
1	0	1
1	1	0

$$\text{NAND} = \begin{bmatrix} 0 & 1 \\ 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

The NOR Gate (NOT-OR)

The output is 1 only when both inputs are 0.

Table 5: Truth Table: NOR Gate

Input A	Input B	Output Y
0	0	1
0	1	0
1	0	0
1	1	0

$$\text{NOR} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}$$

The XOR Gate (Exclusive OR)

The output is 1 if the inputs are different, and 0 if they are identical.

Table 6: Truth Table: XOR Gate

Input A	Input B	Output Y
0	0	0
0	1	1
1	0	1
1	1	0

$$\text{XOR} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

The XNOR Gate (Equivalence Gate)

The inverse of XOR. The output is 1 only if the inputs are identical.

Table 7: Truth Table: XNOR Gate

Input A	Input B	Output Y
0	0	1
0	1	0
1	0	0
1	1	1

$$\text{XNOR} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$